MD. NOMAN BISWAS SIBLY

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EDUCATION

Rajshahi University of Engineering & Technology (RUET) — *Dec 2022 – Present*
B.Sc. in Electrical & Computer Engineering — CGPA: 3.59/4.00 (after 7th semester)

BAF Shaheen College, Jashore — *Jul 2019 – Dec 2021*
HSC – Science — GPA: 5.00/5.00

RESEARCH

Publications

- First Author, "Which Reliability Signal to Trust? Output vs. Representation Space Uncertainty Under Distribution Shift in Pathology Foundation Models", accepted at *MI4MedFM Workshop, MICCAI 2026*. [Code]
- Second Author, "A Comprehensive Review of Convolutional Neural Networks: Foundations, Enhancements, and Applications", *Neural Computing and Applications* (Springer, Q1, IF 5.102), 38(4):56, 2026. [DOI]
- First Author, "Noise-Robust Gallbladder Ultrasound Image Classification Using U-Net GAN Denoising and Vision Transformer", *IEEE QPAIN 2026*. [DOI]

Manuscript in Preparation

- First Author, "JTADC-Net: An Adversarial Joint Task-Aware Denoising and Classification Network for Noise Robust Gallbladder Disease Diagnosis" — joint adversarial denoising and classification for noise-robust gallbladder diagnosis. [Code]

Ongoing Research

- Trustworthy AI in Healthcare (Thesis)

EXPERIENCE

Research Assistant (Part-time, Remote) — 2026 – Present
ELITE Research Lab — *Bangladesh*

· Working on mechanistic interpretability of neural networks to make predictions explainable.

AI & ML Trainee — 2025 – 2026
BDSET — *Bangladesh*

· Six-month program in AI, ML, and immersive technologies under Bangladesh Hi-Tech Park Authority. [Certificate]

Programming Mentor — 2024 – 2025
RUET Analytical Programming Lab — *Rajshahi*

· Taught algorithms/debugging and guided students in competitive programming and contest preparation.

PROJECTS

↪ **JTADC-Net: Noise-Robust Gallbladder Disease Classification:** Developed a GAN-assisted ViT framework for robust gallbladder ultrasound classification under noisy conditions, packaged as a real-time medical image classification web app with deployment support. [Code] [Demo] [2025]

↪ **TruthLayer-Gemma4: Trustworthy Vision-Language AI:** Explored reliability, hallucination reduction, and verification strategies in multimodal vision-language systems, investigating trustworthy AI methods for safer and more interpretable inference. [Code] [2025]

↪ **ATmega Neural Network – Digit Recognizer:** Built a lightweight neural network for digit classification on ATmega microcontrollers using optimized integer computations, with memory-efficient TinyML inference pipelines for constrained embedded environments. [Code] [2025]

PROBLEM SOLVING

Codeforces (Pupil, max 1263; 1186 solved) · **LeetCode** (125 solved) · **VJudge** (378 solved)

CERTIFICATES & ACHIEVEMENTS

6th Place – AI Hackathon, DUET 2026
Dept. of ECE, RUET *Verify*

· Placed 6th at the inter-university AI hackathon hosted at DUET

Machine Learning Specialization (Andrew Ng) 2024
DeepLearning.AI & Stanford *Verify*

· Completed 3-course ML specialization (Supervised ML, Advanced ML, ML Engineering)

CS50: Databases with SQL 2024
Harvard University *Verify*

· Covered SQL, schemas, indexing, transactions, triggers, and database design

CS50: Introduction to Computer Science 2023
Harvard University *Verify*

· Learned Python, algorithms, data structures, and problem-solving

2nd Runner-Up – ECE Youth Programming Contest 2023
RUET

· Ranked 3rd in the department-level competitive programming contest

TECHNICAL SKILLS

↪ **Languages:** C, C++, Python, JavaScript, SQL

↪ **Data Science & ML Tools:** TensorFlow, Scikit-learn, Pandas, NumPy, Matplotlib

↪ **Web Development:** HTML, CSS, Django, MySQL

↪ **Other Tools:** Git, GitHub, Linux, Jupyter Notebook, Data Analysis